Examiner only

6. Nuclear decay in a fission reactor leads to the production of vast amounts of electrical power. The reactor has design features so there is sustained and controlled fission of uranium.

(a) Explain how the reactor is designed so that fission is sustained and controlled. [4]

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(b) A common pair of fission fragments from uranium-235 fission is xenon (Xe) and strontium (Sr). The nuclear equation for the reaction is shown below.

$$^{1}_{0}n + ^{235}_{92}U \rightarrow ^{140}_{54}Xe + ^{94}_{38}Sr + \text{...............}$$

(i) **Complete** the equation. [2]

(ii) Explain why a nucleus of uranium-235 undergoes fission when it absorbs a neutron. [2]

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(c) One product of the fission of uranium is strontium-94. Strontium-94 undergoes beta decay with a half-life of 75 s into an isotope of yttrium (Y). Yttrium also undergoes beta decay and its half-life is 19 minutes. Its product is a stable isotope of zirconium.

(i) Write a balanced nuclear equation for the decay of strontium-94 into yttrium. [2]

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(ii) Explain what is meant by the statement *strontium-94 has a half-life of 75 s.* [2]

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Examiner only

(iii) Explain whether long-term safety precautions are required for the disposal of the waste product strontium-94. [2]

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